

Qualification Pack



Internet of Things (IoT) Assistant

QP Code: NIE/ELE/Q0501

Version: 1.0

NSQF Level: 3

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NIE/ELE/Q0501: Internet of Things (IoT) Assistant

Brief Job Description

The Internet of Things (IoT) Assistant Qualification enable the enrolled trainee identify, assemble and do the basic troubleshooting of various electronic components/product used in the IoT ecosystem. After course completion, the trained learner will exhibit a refined posture about the basic working of different IoT based products, its working, testing, installation, debugging maintenance and troubleshooting.

Personal Attributes

participants should have the ability to analyze and solve problems, ability to deploy IoT solution as per requirement

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ELE/N0513: Identification and troubleshooting of Basic Electronics components](#)
2. [NIE/ELE/N0507: Conceptualising IoT Platform- Arduino](#)
3. [NIE/ELE/N0512: Conceptualising IoT based use-cases](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	IoT Solution, IoT Solution
Country	India
NSQF Level	3
Credits	10
Aligned to NCO/ISCO/ISIC Code	QG-03-IT-00344-2023-V1-NIELIT

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Minimum Educational Qualification & Experience	10th Class (Grade 10th Pass) with NA of experience NA OR 8th Class (Grade 8th Pass with two year of (NTC/ NAC) after 8th Grade 8 pass and pursuing continuous schooling in regular school with vocational subject) with NA of experience OR 8th Class (8th grade pass) with 2 Years of experience
Minimum Level of Education for Training in School	8th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/06/2026
NSQC Approval Date	29/03/2023
Version	1.0
Reference code on NQR	QG-03-IT-00344-2023-V1-NIELIT
NQR Version	1

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NIE/ELE/N0513: Identification and troubleshooting of Basic Electronics components

Description

The NOS covers examine and identification of several electronic devices and components

Scope

The scope covers the following :

- Identification of electronics components
- Understanding the Fundamentals of basic electronics
- Troubleshooting of electronics components
- Understanding the operation of measurement devices

Elements and Performance Criteria

Identification of electronics components

To be competent, the user/individual on the job must be able to:

PC1. Knowledge of basic electrical and electronic components

PC2. able to identify basic electronics components like Resistor, Inductor, diodes, transistor, LED, Capacitor etc.

Fundamentals of basic electronics

To be competent, the user/individual on the job must be able to:

PC3. Knowledge of electronic devices and components

PC4. able to troubleshoot electrical and electronics circuits

Understanding the operation of measurement devices

To be competent, the user/individual on the job must be able to:

PC5. Knowledge of electronic instrumentation devices

PC6. able to work with measuring instruments like multimeter, ammeter, voltmeter, voltage supply, oscilloscope etc

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identification of electronics components</i>	15	7	-	-
PC1. Knowledge of basic electrical and electronic components	-	-	-	-
PC2. able to identify basic electronics components like Resistor, Inductor, diodes, transistor, LED, Capacitor etc.	-	-	-	-
<i>Fundamentals of basic electronics</i>	20	5	-	-
PC3. Knowledge of electronic devices and components	-	-	-	-
PC4. able to troubleshoot electrical and electronics circuits	-	-	-	-
<i>Understanding the operation of measurement devices</i>	15	10	-	-
PC5. Knowledge of electronic instrumentation devices	-	-	-	-
PC6. able to work with measuring instruments like multimeter, ammeter, voltmeter, voltage supply, oscilloscope etc	-	-	-	-
NOS Total	50	22	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0513
NOS Name	Identification and troubleshooting of Basic Electronics components
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	IoT Solution, IoT Solution
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ELE/N0507: Conceptualising IoT Platform- Arduino

Description

The NOS covers IoT development boards, programming and Interfacing

Scope

The scope covers the following :

- Introduction to microprocessor and micro controller
- Introduction to Internet of Things(IoT) applications, protocols, usecases
- Introduction to Arduino
- Embedded C Language
- Interfacing of sensors and actuators with Arduino Boards

Elements and Performance Criteria

Introduction to microprocessor and micro controller

To be competent, the user/individual on the job must be able to:

PC1. Knowledge of microprocessors and microcontrollers

PC2. knowledge of IoT development boards

Introduction to Internet of Things(IoT) –

To be competent, the user/individual on the job must be able to:

PC3. knowledge of IoT architecture, building blocks and use cases

Introduction to C programming

To be competent, the user/individual on the job must be able to:

PC4. Able to write codes in C and Embedded C

Arduino as an IoT development board

To be competent, the user/individual on the job must be able to:

PC5. Knowledge of Arduino Boards and IDE

PC6. interfacing of sensors and actuators on an Arduino board for IoT applications

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to microprocessor and micro controller</i>	20	-	-	-
PC1. Knowledge of microprocessors and microcontrollers	-	-	-	-
PC2. knowledge of IoT development boards	-	-	-	-
<i>Introduction to Internet of Things(IoT) -</i>	20	-	-	-
PC3. knowledge of IoT architecture, building blocks and use cases	-	-	-	-
<i>Introduction to C programming</i>	30	20	10	-
PC4. Able to write codes in C and Embedded C	-	-	-	-
<i>Arduino as an IoT development board</i>	30	25	20	-
PC5. Knowledge of Arduino Boards and IDE	-	-	-	-
PC6. interfacing of sensors and actuators on an Arduino board for IoT applications	-	-	-	-
NOS Total	100	45	30	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0507
NOS Name	Conceptualising IoT Platform- Arduino
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	IoT Solution, IoT Solution
NSQF Level	3
Credits	5
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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NIE/ELE/N0512: Conceptualising IoT based use-cases

Description

The NOS covers various IoT based use case and prototype development

Scope

The scope covers the following :

- Smart Street Light control
- Home automation using Arduino
- Password enabled Digital Lock using Arduino

Elements and Performance Criteria

Smart Street Light control

To be competent, the user/individual on the job must be able to:

PC1. IoT based prototype development for Smart Street Light control

Home automation using Arduino

To be competent, the user/individual on the job must be able to:

PC2. Development of an home automation system

Password enabled Digital Lock using Arduino

To be competent, the user/individual on the job must be able to:

PC3. Able to develop Digital Locks using Arduino

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Smart Street Light control</i>	15	7	-	-
PC1. IoT based prototype development for Smart Street Light control	-	-	-	-
<i>Home automation using Arduino</i>	20	8	-	-
PC2. Development of an home automation system	-	-	-	-
<i>Password enabled Digital Lock using Arduino</i>	15	8	-	-
PC3. Able to develop Digital Locks using Arduino	-	-	-	-
NOS Total	50	23	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0512
NOS Name	Conceptualising IoT based use-cases
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	IoT Solution, IoT Solution
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	29/03/2023
Next Review Date	29/03/2026
NSQC Clearance Date	29/03/2023

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2026
Next Review Date	18/02/2029
NSQC Clearance Date	18/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.

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6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0513.Identification and troubleshooting of Basic Electronics components	50	22	-	-	72	20
NIE/ELE/N0507.Conceptualising IoT Platform- Arduino	100	45	30	-	175	51
NIE/ELE/N0512.Conceptualising IoT based use-cases	50	23	-	-	73	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	9
Total	220	120	30	-	370	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.